

Nikunj Pradhan

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EDUCATION

Missouri University of Science and Technology

Masters of Computer Science

Bachelor of Science: Computer Science

GPA: 4.0/4.0

Aug 2025 - May 2027

Aug 2023 - Dec 2026

EXPERIENCE

Data Engineer Co-Op

Hunter Engineering Company

Jan 2025 – Aug 2025

Bridgeton, MO

- Engineered real-time production model dashboards, data pipelines, dataset analytical platform, and AI-assisted annotation endpoints.
- Automated benchmarking pipelines across cloud platforms with serverless functions (Lambda, Azure Functions) to compare performance of in-house and third-party AI/ML models.
- Accelerated distributed multi-GPU training workflows by 66% and development-to-production cycles through close collaboration with R&D teams.
- Created and maintained custom ETL pipelines using FastAPI and Python to clean, validate, and process metadata and image data from 600+ worksites, ingesting over 2M images into Microsoft SQL Server and AWS S3.
- Built a CNN-based regression model predicting object coordinates, improving geometric correction accuracy by 23%.

Software Engineer

Tally Receipts

June 2025 – July 2025

St. Louis, MO

- Redesigned and migrated backend from MongoDB to AWS DynamoDB, redesigning schemas for efficient querying, scalability, and data lifecycle management.
- Built an LLM-powered receipt categorization service using LlamaIndex for receipt summarization and classification, integrating with calendar workflows to reduce manual entry effort for users.
- Implemented backend logic for secure card-based transaction linking, transaction tracking, and webhook processing to collect and manage user receipts.

Research Assistant

Missouri University of Science and Technology

Aug 2023 – Dec 2024

Rolla, MO

- Improved CV model accuracy by 62% through optimization, hyperparameter tuning, and algorithmic refinement using C++ OpenCV and Point Cloud.
- Developed a MATLAB/C++ application simulating 3D triaxial shear tests with deep learning and parallel computing for soil compression modeling, incorporating advanced CV techniques like watershed segmentation and 3D model triangulation via ray tracing.

PROJECTS

Devanagari OCR Pipeline Application | *FastAPI, Python, Pytorch, NLTK, MLOps*

- Deployed a full Nepali OCR pipeline +130 characters CRNN text recognizer with integrated real-time text-detection trained on 500k+ samples and delivered as a FastAPI inference workflow API service.

Movie Recommendation & Watchlist Platform | *FastAPI, React.js, PostgreSQL, MinIO, Scikit-learn*

- Developed and hosted full-stack platform featuring secure 2FA, personalized watchlists, and a hybrid recommendation engine (content & collaborative) with per-user caching, deployed on a scalable Kubernetes cluster with NGINX and MinIO.

Real-Time BlackJack Statistical Engine | *FastAPI, Apache Kafka, Numpy*

- Architected a high-throughput decision engine integrating a CV pipeline and Apache Kafka to stream real-time data into a NumPy-vectorized Monte Carlo engine, simulating 10^5 concurrent states to calculate optimal Expected Value (EV).

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL, Java, HTML/CSS

Web Development & APIs: FastAPI, Spring Boot, React, Django, Node.js, WebSockets, Apache Kafka

Cloud & DevOps: AWS (S3, EC2, Lambda), Docker, Kubernetes(k8s), Git, CI/CD, Linux/Ubuntu, CUDA

Databases: PostgreSQL, Pgvector, Microsoft SQL, DynamoDB, MinIO, MongoDB

Machine Learning & AI: Pytorch, Scikit-learn, HuggingFace, XGBoost, LlamaIndex, LangChain

Data Science Tools: Pandas, NumPy, SciPy, Matplotlib, Jupyter, OpenCV, NLTK, Redis, Playwright, BeautifulSoup